

EXPERIMENT 11 : Simple Top-Down Parser for Context-Free Grammar

Aim: To implement a simple top-down parser for a Context-Free Grammar (CFG) using Python.

Program:

```
grammar = {  
    "S": ["NP", "VP"],  
    "NP": ["Det", "N"],  
    "VP": ["V", "NP"],  
    "Det": ["the", "a"],  
    "N": ["student", "teacher", "book"],  
    "V": ["reads", "likes"]  
}  
  
def parse(symbol, words, index):  
    if symbol not in grammar:  
        if index < len(words) and symbol == words[index]:  
            return index + 1  
        return None  
    for rule in grammar[symbol]:  
        current = index  
        success = True  
        for item in rule:  
            current = parse(item, words, current)  
            if current is None:  
                success = False  
                break  
        if success:  
            return current  
    return None
```

```

sentence = input("Enter a sentence: ").lower()

words = sentence.split()

result = parse("S", words, 0)

if result == len(words):

    print("Sentence is accepted by the grammar.")

else:

    print("Sentence is rejected by the grammar.")

```

Output:

The screenshot shows a Python IDE with two panes. The left pane displays the source code for a file named `EXP11.py`. The code defines a grammar and a `parse` function. The grammar rules are: `S` is `NP VP`, `NP` is `Det N`, `VP` is `vp NP`, `Det` is `the a`, `N` is `student teacher book`, and `vp` is `reads likes`. The `parse` function checks if a given symbol can be derived from the grammar starting from `S`. The main code prompts the user to enter a sentence, splits it into words, and calls `parse("S", words, 0)`. If the result equals the number of words, it prints "Sentence is accepted by the grammar."; otherwise, it prints "Sentence is rejected by the grammar.".

The right pane shows the IDLE Shell output. It displays the Python version (3.14.0), the file path, and the execution of the program. The user input "the student reads a book" is shown, followed by the output "Sentence is accepted by the grammar.".

```

EXP11.py - C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP11.py (3.14.0)
File Edit Format Run Options Window Help

grammar = {
    "S": [{"NP", "vp"}],
    "NP": [{"Det", "N"}],
    "vp": [{"vp", "NP"}],
    "Det": [{"the", "a"}],
    "N": [{"student", "teacher", "book"}],
    "vp": [{"reads", "likes"}]
}

def parse(symbol, words, index):
    if symbol not in grammar:
        if index < len(words) and symbol == words[index]:
            return index + 1
        return None
    for rule in grammar[symbol]:
        current = index
        success = True
        for item in rule:
            current = parse(item, words, current)
            if current is None:
                success = False
                break
        if success:
            return current
    return None

sentence = input("Enter a sentence: ").lower()
words = sentence.split()
result = parse("S", words, 0)
if result == len(words):
    print("Sentence is accepted by the grammar.")
else:
    print("Sentence is rejected by the grammar.")

Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP11.py
Enter a sentence: the student reads a book
Sentence is accepted by the grammar.
>>>

```